

Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised
Learning

K-means Clustering
Principal Component
Analysis (PCA)

Introduction to Machine Learning and High Performance Computing

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Supervised vs.
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1 Basic Concepts Supervised vs. Unsupervised Learning

2 Unsupervised Learning K-means Clustering Principal Component Analysis (PCA)

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Learning

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Principal Component
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Supervised Learning (learning with answers)

- The computer is trained using examples that already have the right answers.
- **Example:** You show a program many pictures labeled “dog” or “cat” so it learns to tell them apart.
- It gets feedback — we can check if its guesses are correct or not.

Unsupervised Learning (learning without answers)

- The computer only sees the raw data and must find patterns on its own.
- **Example:** You give the program a folder of animal photos with no labels, and it groups them by looks or size.
- No one tells it if it's right or wrong — it just discovers what's similar.

Which data are labelled?

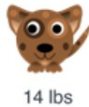
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Supervised vs.
Unsupervised Learning

Unsupervised Learning

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Principal Component
Analysis (PCA)



Question

Which ones represent **labelled data**, and which shows **unlabelled data**?

Which data are labelled?

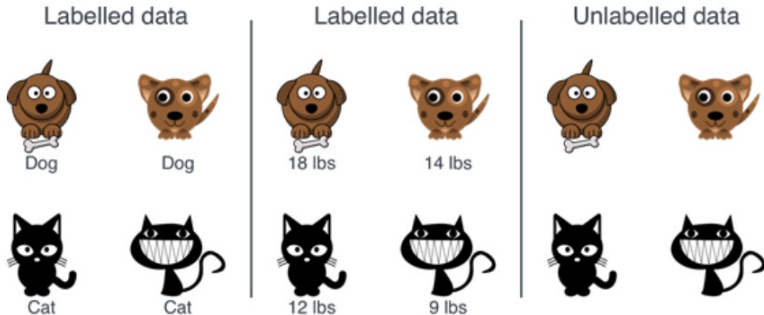
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Supervised vs.
Unsupervised Learning

Unsupervised
Learning

K-means Clustering
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Analysis (PCA)



Kudos! How many of you got it right?

The two on the left are **labelled data** — they already have answers.
On the right, we have **unlabelled data** — the computer figures out the patterns.

Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

1 Basic Concepts Supervised vs. Unsupervised Learning

2 Unsupervised Learning K-means Clustering Principal Component Analysis (PCA)

Unsupervised Learning

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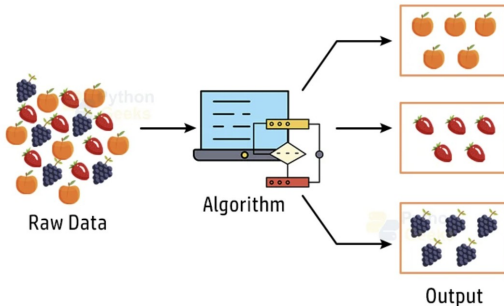
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Supervised vs.
Unsupervised Learning

Unsupervised Learning

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- **Input:** Features X only (no labels or outcomes y)
- **Clustering:**
 - Group similar things together so that items in the same group are alike, and those in different groups are different.
 - Useful for finding structure or patterns in a dataset.



Classification vs. Clustering

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Unsupervised Learning

Unsupervised Learning

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Clustering: The labels are unknown — the algorithm groups similar data automatically.

Clustering



C 1

C 2

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Classification: We already know the labels — the model learns to assign new data to one of them.

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Class 1



Class 3



Class 2



Class 4

Classification

Quick Quiz: Clustering or Classification?

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Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

- 1 Your phone sorts photos of your friends automatically.

Quick Quiz: Clustering or Classification?

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Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

- 1 Your phone sorts photos of your friends automatically.
- 2 Gmail filters messages into *Primary*, *Social*, or *Promotions*.

Clustering

Quick Quiz: Clustering or Classification?

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Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

- ① Your phone sorts photos of your friends automatically.
- ② Gmail filters messages into *Primary*, *Social*, or *Promotions*.
- ③ Spotify groups songs you might like into a new playlist.

Clustering
Classification

Quick Quiz: Clustering or Classification?

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Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

- ① Your phone sorts photos of your friends automatically. Clustering
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Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

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Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
Principal Component
Analysis (PCA)

Applications:

- **Chemistry:** Grouping chemical compounds by structural similarity.
- **Molecular & Cell Biology:** Identifying cell types from gene expression data.
- **Bioengineering:** Segmenting patients based on physiological signals.
- **Engineering:** Image segmentation, user behavior clustering, or document topic modeling.

K-Means Clustering

K-Means Clustering

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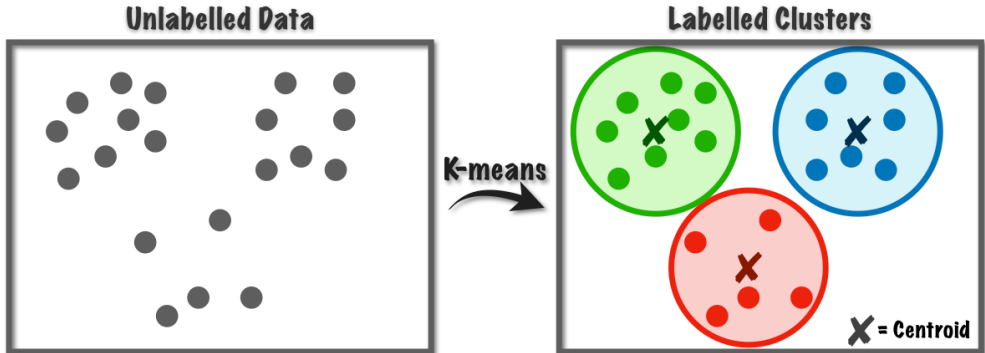
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Supervised vs.
Unsupervised Learning

Unsupervised Learning

K-means Clustering
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- One of the simplest and most popular clustering algorithms.
- You choose the number of clusters, K .
- The algorithm assigns each data point to the nearest cluster center (centroid).
- Then, it updates the centers and repeats until the groups stabilize.



K-Means Algorithm

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Supervised vs.
Unsupervised Learning

Unsupervised Learning

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Principal Component
Analysis (PCA)

Goal: Partition n observations into K clusters by minimizing the **within-cluster sum of squares (WCSS)**:

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} [\text{distance}(x_i, \mu_k)]^2 \quad ; \quad \mu_k = \text{center of cluster } C_k$$

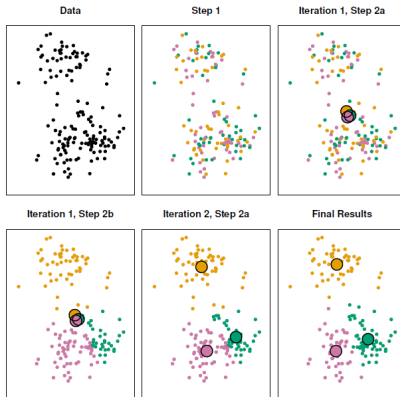
Algorithm steps:

- ① **Start:** Pick K random initial centroids.
- ② **Assign:** Each data point joins the nearest centroid.
- ③ **Update:** Recalculate centroids as the average of points assigned to each cluster.
- ④ **Repeat:** Repeat 2 & 3 until cluster assignments stop changing.

Example: K-Means in Action

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K-Means algorithm on an example with $K = 3$:



Top row: Random start, first assignments, and initial centroid updates.

Bottom row: Reassign points, update again, and reach stable clusters.

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K-means Clustering
Principal Component
Analysis (PCA)

K-Means Considerations

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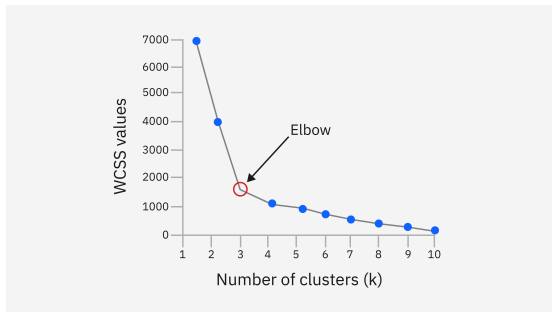
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Supervised vs.
Unsupervised Learning

Unsupervised Learning

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Principal Component
Analysis (PCA)

- **Initial centroid assignment matters:** K-Means can produce different results depending on how the first centroids are chosen. Running it multiple times helps find a more reliable solution.
- **How many clusters?** Try several values of K and look for the "elbow" — the point where adding more clusters stops improving much.



The **Elbow Method**: Find the point where the curve bends — that's a good choice for K .

Principal Component Analysis (PCA)

Dimensionality Reduction

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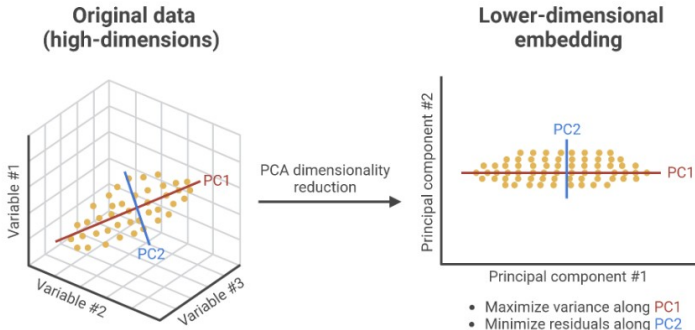
Unsupervised Learning

K-means Clustering
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Analysis (PCA)

Goal: Make high-dimensional data simpler while keeping most of its important variation.

How It Works:

- Finds new directions that show the biggest differences in the data.
- Creates new, uncorrelated features called **principal components (PCs)**.
- Turns many related features into just a few key ones (e.g., 2D or 3D).



Why and Where We Use PCA

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Unsupervised Learning

Unsupervised Learning

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Principal Component
Analysis (PCA)

Why Use PCA?

- Simplifies large, complex datasets.
- Removes overlap and redundancy between features.
- Helps reveal hidden patterns and relationships.
- Speeds up other ML models.

Real-World Applications:

- **Chemistry:** Visualizing complex chemical spectra in 2D.
- **Molecular Biology:** Reducing gene expression data to see cell groups.
- **Bioengineering:** Simplifying signals from multiple sensors.
- **Finance:** Finding patterns in stock movements or risk factors.

In Simple Terms

PCA turns “too many variables” into “just the ones that matter.”

Quiz: When Would You Use PCA?

Which of these situations is a good use case for PCA?

- ① You want to group similar playlists on a music app based on listening habits.
- ② You have hundreds of daily measurements from fitness trackers and want to uncover overall activity patterns.
- ③ You want to predict if a product review is positive or negative.
- ④ You're analyzing hundreds of financial indicators and want to find the few main market trends.

Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

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Principal Component
Analysis (PCA)

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Supervised vs.
Unsupervised LearningUnsupervised
LearningK-means Clustering
Principal Component
Analysis (PCA)

Quiz: When Would You Use PCA?

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- ③ You want to predict if a product review is positive or negative.
- ④ You're analyzing hundreds of financial indicators and want to find the few main market trends.

Answer: (2) and (4) — PCA helps when you have many related variables and want to focus on the main sources of variation in your data.

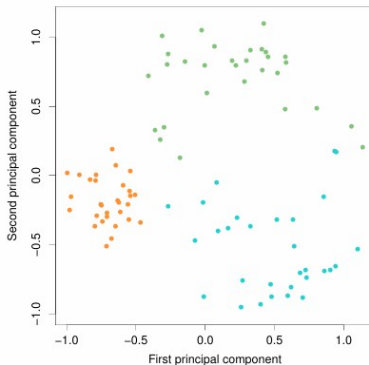
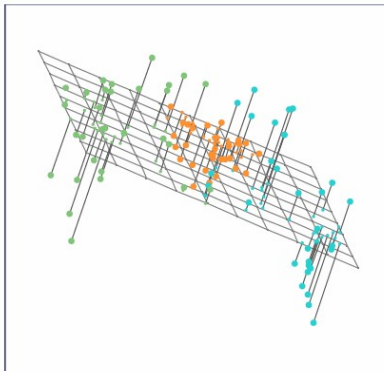
Input: p correlated features $X_1, \dots, X_p$.

- ① **Standardize the data:** Make all features comparable by removing the mean and scaling to unit variance.
- ② **Look for relationships:** Compute how variables move together — the *covariance matrix*.
- ③ **Find main directions of variation:** Use the *eigenvectors* of that matrix — these become the new axes (PC1, PC2, ...).
- ④ **Re-express the data:** Project everything onto these new axes to get fewer, uncorrelated features.
- ⑤ **Keep only the important ones:** Select the first few PCs that explain most of the total variation.

Example: PCA in Action

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90 observations simulated in three dimensions (color is for ease of visualization)



Left: 3D data points with a plane (PC1–PC2) that best fits the data.

Right: Projection of the same points onto that plane (the 2D PCA view).

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Unsupervised
Learning

K-means Clustering

Principal Component
Analysis (PCA)

PCA Considerations

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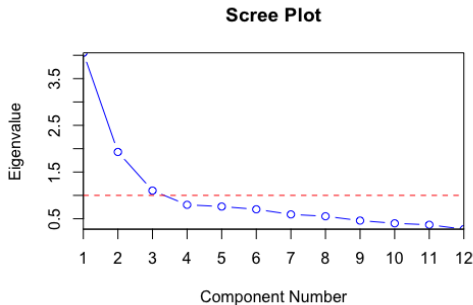
Basic Concepts

Supervised vs.
Unsupervised Learning

Unsupervised Learning

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Analysis (PCA)

- **Explained variance:** The first few PCs capture most of the variation (NOT ALL). Later ones add little new information.
- **Interpretation:** PCs blend the original features into new axes that capture variation but may not have an obvious real-world meaning.
- **Choosing how many PCs:** Use a **scree plot** or **cumulative variance plot**. Look for the “*elbow*” — where adding more PCs gives small gains.



The elbow marks where additional PCs no longer add much information.